EC301 — Electronic Devices

Semester III • 3L:0T:0P • 3 Credits

Module 2 Exam Notes: P-N Junction & Bipolar Junction Transistor

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1 Physical Description of the P-N Junction **2**
1.1 Thermal Equilibrium and Depletion Region Formation 2
1.2 Derivation of the Built-In Potential (V_{bi}) 3
1.3 Depletion Width and Electric Field Analysis 3
1.4 Energy Band Diagrams under Bias 4
2 P-N Junction Fabrication Technologies **4**
3 I-V Characteristics and Small Signal Switching Models **5**
3.1 The Shockley Diode Equation 5
3.2 Small-Signal AC Model of P-N Diode 6
3.3 Diode Switching Characteristics and Reverse Recovery Time 6
4 Junction Breakdown Mechanisms and Diodes **6**
4.1 Zener Breakdown vs. Avalanche Breakdown 7
4.2 Schottky Diode (Metal-Semiconductor Junction) 7
5 Bipolar Junction Transistor (BJT) **8**
5.1 Basic Construction and Doping Asymmetry 8
5.2 BJT Operating Modes 8
5.3 Current Relationships in Forward Active Mode 8
5.4 Base Width Modulation (The Early Effect) 9
5.5 BJT I-V Characteristics (Common-Emitter Input and Output) 9
6 The Ebers-Moll Model **10**
6.1 Model Formulation and Equivalent Circuit 10

1. Physical Description of the P-N Junction

A **p-n junction** is formed when a p-type semiconductor material is joined seamlessly at an atomic level with an n-type semiconductor material within a single continuous crystal lattice.

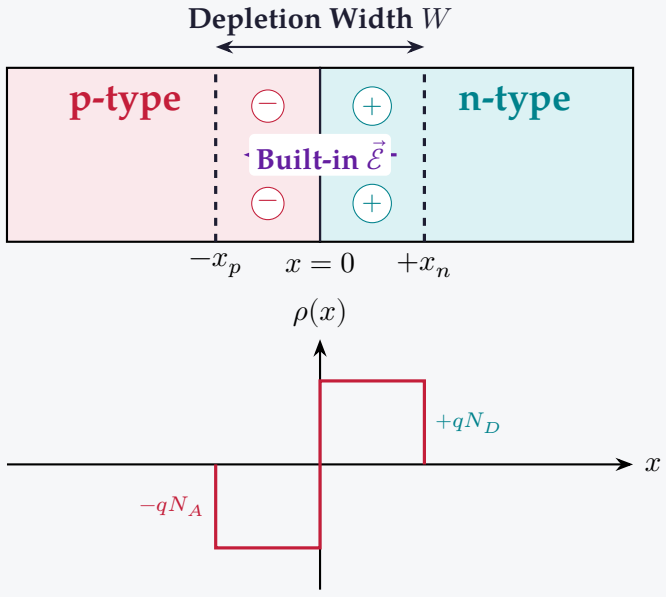
1.1 Thermal Equilibrium and Depletion Region Formation

Due to the steep concentration gradients across the metallurgical junction ($x = 0$):

- Majority holes in the p-region diffuse across the interface into the n-region, where they recombine with majority electrons.
- Majority electrons in the n-region diffuse into the p-region and recombine with holes.
- This diffusion uncovers uncompensated, immobile ionized dopant cores:
  - Negatively charged acceptor ions (N_A^-) in the p-side ($-x_p \leq x \leq 0$).
  - Positively charged donor ions (N_D^+) in the n-side ($0 \leq x \leq x_n$).
- The region containing these uncovered unneutralized fixed charges is devoid of mobile carriers and is called the **depletion region** (or **space charge region**).
- The fixed charges establish an internal electric field $\vec{\mathcal{E}}$ pointing from the positive donor ions ($+x$) to the negative acceptor ions ($-x$). This field exerts a drift force that opposes further carrier diffusion.
- At **thermal equilibrium**, the drift current driven by $\vec{\mathcal{E}}$ exactly balances the diffusion current driven by concentration gradients, resulting in zero net current:

$$J_n = J_{n,\text{drift}} + J_{n,\text{diff}} = 0, \qquad J_p = J_{p,\text{drift}} + J_{p,\text{diff}} = 0 \tag{1}$$

Space Charge Region and Charge Density Profile of an Abrupt P-N Junction



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1.2 Derivation of the Built-In Potential (V_{bi})

Under thermal equilibrium, setting the hole current density to zero:

$$J_p = qp\mu_p\mathcal{E} - qD_p\frac{dp}{dx} = 0 \tag{2}$$

$$\mu_p\mathcal{E} = D_p\frac{1}{p}\frac{dp}{dx} \tag{3}$$

Using the Einstein relation $\frac{D_p}{\mu_p} = \frac{kT}{q} = V_T$ and the electrostatic relation $\mathcal{E} = -\frac{dV}{dx}$:

$$-\frac{dV}{dx} = \frac{kT}{q}\frac{1}{p}\frac{dp}{dx} \implies -dV = \frac{kT}{q}\frac{dp}{p} \tag{4}$$

Integrating from the neutral p-region ($x = -x_p$, where $p = p_{p0} \approx N_A$ and $V = 0$) to the neutral n-region ($x = x_n$, where $p = p_{n0} = n_i^2/N_D$ and $V = V_{bi}$):

$$-\int_0^{V_{bi}} dV = \frac{kT}{q}\int_{p_{p0}}^{p_{n0}}\frac{dp}{p} \tag{5}$$

$$-V_{bi} = \frac{kT}{q}\ln\left(\frac{p_{n0}}{p_{p0}}\right) = -\frac{kT}{q}\ln\left(\frac{p_{p0}}{p_{n0}}\right) \tag{6}$$

Substituting $p_{p0} \approx N_A$ and $p_{n0} = \frac{n_i^2}{N_D}$:

Built-In Potential (Contact Potential)

$$V_{bi} = V_T\ln\left(\frac{N_AN_D}{n_i^2}\right) = \frac{kT}{q}\ln\left(\frac{N_AN_D}{n_i^2}\right) \tag{7}$$

At $T = 300\,\text{K}$ ($V_T \approx 25.9\,\text{mV}$), for Silicon with $N_A = 10^{16}\,\text{cm}^{-3}$, $N_D = 10^{17}\,\text{cm}^{-3}$: $V_{bi} \approx 0.0259\ln\left(\frac{10^{33}}{2.25\times10^{20}}\right) \approx 0.75\,\text{V}$.

1.3 Depletion Width and Electric Field Analysis

By integrating Poisson's equation $\frac{d\mathcal{E}}{dx} = \frac{\rho(x)}{\epsilon_s}$ across an abrupt step junction:

- For $-x_p \le x \le 0$: $\mathcal{E}(x) = -\frac{qN_A}{\epsilon_s}(x + x_p)$
- For $0 \le x \le x_n$: $\mathcal{E}(x) = +\frac{qN_D}{\epsilon_s}(x - x_n)$

By overall charge neutrality, the total negative space charge must balance the positive space charge:

$$qN_Ax_p = qN_Dx_n \implies \frac{x_n}{x_p} = \frac{N_A}{N_D} \tag{8}$$

The depletion region penetrates deeper into the more lightly doped semiconductor side.

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Depletion Width and Peak Electric Field under Applied Bias V_a

For an applied voltage V_a ($V_a > 0$ for forward bias, $V_a = -V_R$ for reverse bias):

$$W = x_n + x_p = \sqrt{\frac{2\epsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)(V_{bi} - V_a)} \tag{9}$$

Individual depletion layer penetrations:

$$x_n = W\left(\frac{N_A}{N_A + N_D}\right), \qquad x_p = W\left(\frac{N_D}{N_A + N_D}\right) \tag{10}$$

The maximum electric field occurs at the metallurgical junction ($x = 0$):

$$\mathcal{E}_{\max} = -\frac{2(V_{bi} - V_a)}{W} = -\frac{qN_D x_n}{\epsilon_s} = -\frac{qN_A x_p}{\epsilon_s} \tag{11}$$

1.4 Energy Band Diagrams under Bias

- **Thermal Equilibrium ($V_a = 0$):** The Fermi level is flat throughout the entire device ($dE_F/dx = 0$). The total band bending is equal to the built-in potential barrier: qV_{bi}.
- **Forward Bias ($V_a > 0$):** Positive terminal connected to p-side, negative to n-side. The potential barrier is reduced to $q(V_{bi} - V_a)$. Depletion width narrows, and the Fermi level splits into two separate **quasi-Fermi levels**: $E_{Fn} - E_{Fp} = qV_a$. Majority carriers readily diffuse across the lowered barrier.
- **Reverse Bias ($V_a = -V_R < 0$):** Positive terminal connected to n-side, negative to p-side. The potential barrier increases to $q(V_{bi} + V_R)$. Depletion width expands, completely blocking majority diffusion. Only thermally generated minority carriers drift across the junction, producing a tiny saturation current I_0.

2. P-N Junction Fabrication Technologies

Modern semiconductor devices are fabricated using planar integrated circuit technologies:

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Method	Process Description	Key Characteristics
Grown Junction	Dopant impurities are introduced into the molten melt during crystal pulling (Czochralski process).	Historical method; poor dimensional control; abruptness limited.
Alloyed Junction	A pellet of trivalent impurity (e.g., Indium) is melted onto an n-type substrate and recrystallized.	Forms extremely abrupt junctions; rugged; large junction depth variations.
Thermal Diffusion	Dopant vapor is diffused at elevated temperatures (900–1200°C) into masked wafer windows.	Planar geometry; excellent reproducibility; lateral diffusion under oxide masks occurs.
Ion Implantation	Dopant ions are ionized, accelerated to high energies (10–400 keV), and fired into the wafer.	Precise dosage and depth profile control; low processing temperature; requires thermal annealing to heal crystal damage.
Epitaxial Growth	High-purity single-crystal layer grown on single-crystal substrate using Chemical Vapor Deposition (CVD).	Controlled uniform thickness and doping; used for collector layers and buried layers.

3. I-V Characteristics and Small Signal Switching Models

3.1 The Shockley Diode Equation

Under the assumption of low-level injection and steady-state conditions, the current flowing through a p-n junction diode is given by the **Shockley Ideal Diode Equation**:

Shockley Diode Equation and Reverse Saturation Current

$$I = I_0 \left[\exp\left(\frac{qV_a}{\eta kT}\right) - 1\right] = I_0 \left[\exp\left(\frac{V_a}{\eta V_T}\right) - 1\right] \tag{12}$$

where η is the **ideality factor** ($\eta \approx 1$ for Germanium; $\eta \approx 2$ at low current in Silicon where depletion recombination dominates, and $\eta \approx 1$ at higher forward currents where diffusion dominates).
The **reverse saturation current** I_0 is:

$$I_0 = qA\left(\frac{D_p p_{n0}}{L_p} + \frac{D_n n_{p0}}{L_n}\right) = qAn_i^2\left(\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A}\right) \tag{13}$$

Because $I_0 \propto n_i^2 \propto T^3 \exp(-E_g/kT)$, I_0 is extremely temperature-sensitive:
- I_0 approximately **doubles for every** 10°**C rise** in temperature for Silicon.
- Forward voltage drop decreases by approximately $-$**2** mV/°C for a constant forward current.

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3.2 Small-Signal AC Model of P-N Diode

When a small sinusoidal AC signal ($v_d \ll V_T$) is superimposed on a DC operating point (V_D, I_D):

1. **Dynamic (AC) Resistance (r_d):** The slope of the forward I-V characteristic:

$$r_d = \left(\frac{dI}{dV}\right)^{-1} = \frac{\eta V_T}{I_D + I_0} \approx \frac{\eta V_T}{I_D} \tag{14}$$

   At $I_D = 1\,\text{mA}$, for $\eta = 1$: $r_d \approx \frac{25.9\,\text{mV}}{1\,\text{mA}} \approx 26\,\Omega$.

2. **Junction (Depletion) Capacitance (C_j):** Arises from dipole charge variation (dQ/dV) stored in the uncompensated depletion region under reverse bias:

$$C_j = \frac{\epsilon_s A}{W} = \frac{C_{j0}}{\left(1 + \frac{V_R}{V_{bi}}\right)^m} \tag{15}$$

   where $m = 1/2$ for an abrupt (step) junction, and $m = 1/3$ for a linearly graded junction. Dominates under **reverse bias**.

3. **Diffusion Capacitance (C_d):** Arises from the rearrangement of injected excess minority carrier charge stored in the neutral regions under forward bias:

$$C_d = \frac{dQ_{\text{diff}}}{dV_a} = \frac{d(I_D \tau_T)}{dV_a} = \tau_T \frac{dI_D}{dV_a} = \frac{\tau_T I_D}{\eta V_T} \tag{16}$$

   where τ_T is the minority carrier transit time. C_d is directly proportional to forward current I_D and dominates heavily under **forward bias**.

3.3 Diode Switching Characteristics and Reverse Recovery Time

When a diode is switched rapidly from forward conduction (I_F) to reverse bias ($-V_R$):

- The diode cannot turn off instantaneously because a substantial amount of excess minority carrier charge $Q_s = I_F \tau_T$ is stored in the neutral base.
- **Storage Time (t_s):** The time required to extract the stored excess charge near the junction until the carrier density drops to equilibrium. Current remains high and negative ($I_R \approx -V_R/R_L$).
- **Transition Time (t_t):** The time required for the junction depletion layer capacitance to charge up to the reverse bias voltage $-V_R$. Current decays asymptotically to $-I_0$.
- **Reverse Recovery Time (t_{rr}):** Total turn-off time:

$$t_{rr} = t_s + t_t \tag{17}$$

4. Junction Breakdown Mechanisms and Diodes

When a sufficiently large reverse bias voltage ($V_R = V_{BR}$) is applied across a p-n junction, the reverse current increases drastically with negligible increase in voltage.

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4.1 Zener Breakdown vs. Avalanche Breakdown

Parameter	Zener Breakdown	Avalanche Breakdown
Doping Concentration	Heavily doped on both p and n sides ($N_A, N_D > 10^{18}\,\text{cm}^{-3}$)	Lightly to moderately doped ($N_A, N_D < 10^{16}\,\text{cm}^{-3}$)
Depletion Width (W)	Extremely thin ($W < 10\,\text{nm} = 100\,\text{Å}$)	Relatively wide ($W > 1\,\mu\text{m}$)
Breakdown Voltage (V_{BR})	Occurs at low reverse voltages ($V_{BR} < 4$–$6\,\text{V}$)	Occurs at higher reverse voltages ($V_{BR} > 6\,\text{V}$)
Physical Mechanism	Direct quantum mechanical tunneling of valence electrons from p-side VB to n-side CB across thin barrier	Impact ionization: high kinetic energy carriers collide with lattice atoms, generating secondary EHPs (avalanche multiplication)
Electric Field	High electric field ($\mathcal{E} \geq 10^6\,\text{V/cm}$)	Moderate to high field ($\mathcal{E} \sim 2$–$4 \times 10^5\,\text{V/cm}$)
Temperature Coefficient	**Negative:** As T rises, bandgap E_g narrows, making tunneling easier $\implies$ V_Z decreases	**Positive:** As T rises, lattice vibrations increase, shortening mean free path, requiring higher voltage for collision ionization $\implies$ V_A increases

4.2 Schottky Diode (Metal-Semiconductor Junction)

A **Schottky barrier diode** is formed by depositing a metal (e.g., Gold, Platinum, Tungsten, Aluminum) directly onto an n-type semiconductor.

- **Schottky Barrier Height (Φ_B):**
$$\Phi_B = \Phi_m - \chi_s \tag{18}$$
where Φ_m is the metal work function, and χ_s is the semiconductor electron affinity.
- **Built-in Potential (V_{bi}):**
$$V_{bi} = \Phi_B - (E_c - E_F)/q \tag{19}$$
- **Majority Carrier Conduction:** Current conduction occurs entirely via majority electron **thermionic emission** over the barrier into the metal.
- **Zero Minority Storage Time:** Because there is virtually no minority carrier injection into the semiconductor, there is zero stored charge ($Q_s \approx 0$). The reverse recovery time t_{rr} is virtually zero ($\sim$ 10–100 ps), making Schottky diodes ideal for high-speed switching and ultra-high frequency (RF/microwave) clamping circuits.
- **Low Cut-In Voltage:** The forward knee voltage is only 0.2–0.3 V (compared to 0.7 V for Silicon p-n diodes).

5. Bipolar Junction Transistor (BJT)

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The Bipolar Junction Transistor is a three-terminal, two-junction semiconductor device whose operation depends on both majority and minority carrier transport (hence **bipolar**).

5.1 Basic Construction and Doping Asymmetry

A BJT consists of three alternating regions: Emitter (E), Base (B), and Collector (C), forming an n-p-n or p-n-p sandwich:

Region	Doping Concentration	Physical Width	Primary Functional Role
Emitter (E)	Heavily doped ($N_E \approx 10^{19}\,\text{cm}^{-3}$)	Moderate	Injects majority carriers into the base with maximum injection efficiency.
Base (B)	Very lightly doped ($N_B \approx 10^{16}\,\text{cm}^{-3}$)	Extremely narrow ($W_b \ll L_{\text{minority}}$)	Transports injected minority carriers to the collector with minimum recombination.
Collector (C)	Moderately / Lightly doped ($N_C \approx 10^{15}\,\text{cm}^{-3}$)	Largest physical dimensions	Collects carriers sweeping across reverse-biased CB junction; dissipates heat.

5.2 BJT Operating Modes

Operating Mode	Emitter-Base Junction (EBJ)	Collector-Base Junction (CBJ)	Primary Application
Forward Active	Forward Biased	Reverse Biased	Linear analog amplification
Cut-off	Reverse Biased	Reverse Biased	Digital logic "OFF" switch ($I_C \approx 0$)
Saturation	Forward Biased	Forward Biased	Digital logic "ON" switch ($V_{CE,\text{sat}} \approx 0.2\,\text{V}$)
Inverse / Inverted	Reverse Biased	Forward Biased	Specialized bidirectional switching

5.3 Current Relationships in Forward Active Mode

For an n-p-n transistor in forward active mode:

1. Electrons injected from Emitter into Base form emitter current I_E.
2. A tiny fraction ($< 1\%$) recombines with holes in the thin base, forming base current I_B.
3. The remaining vast majority ($> 99\%$) diffuses across the neutral base, reaches the CB depletion edge, and is swept into the collector by the strong reverse electric field, forming collector current I_C:

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BJT Current Relationships

$$I_E = I_B + I_C \tag{20}$$

Common-Base Current Gain (α):

$$\alpha = \frac{I_C}{I_E} \approx 0.95 \text{ to } 0.998 \tag{21}$$

Common-Emitter Current Gain (β or h_{FE}):

$$\beta = \frac{I_C}{I_B} = \frac{\alpha}{1-\alpha} \approx 50 \text{ to } 500 \tag{22}$$

In terms of reverse collector saturation current (I_{CBO}):

$$I_C = \alpha I_E + I_{CBO} = \beta I_B + (\beta+1) I_{CBO} = \beta I_B + I_{CEO} \tag{23}$$

5.4 Base Width Modulation (The Early Effect)

In forward active mode, as the reverse bias voltage across the collector-base junction (V_{CB} or V_{CE}) increases:

- The collector-base space charge region expands into the lightly doped base.
- The **effective neutral base width** ($W_{\text{eff}} = W_{\text{base}} - x_{p,\text{dep}}$) decreases.
- **Consequences**:
  1. Less recombination in the base $\implies$ Base transport factor increases $\implies$ α and β increase.
  2. The concentration gradient of minority electrons in the base ($dn/dx \approx n(0)/W_{\text{eff}}$) steepens $\implies$ I_C increases with increasing V_{CE}.
- When extrapolated backward, the linear output characteristics in the active region intersect the negative voltage axis at a common voltage called the **Early Voltage** ($V_A \approx$ 50–100 V).
- The collector current incorporating the Early effect is modeled as:

$$I_C = I_S \exp\left(\frac{V_{BE}}{V_T}\right)\left(1 + \frac{V_{CE}}{V_A}\right) \tag{24}$$

yielding a finite small-signal output resistance: $r_o = \frac{V_A + V_{CE}}{I_C} \approx \frac{V_A}{I_C}$.

5.5 BJT I-V Characteristics (Common-Emitter Input and Output)

The static electrical behavior of a BJT in the Common-Emitter (CE) configuration is described by two sets of characteristic curves:

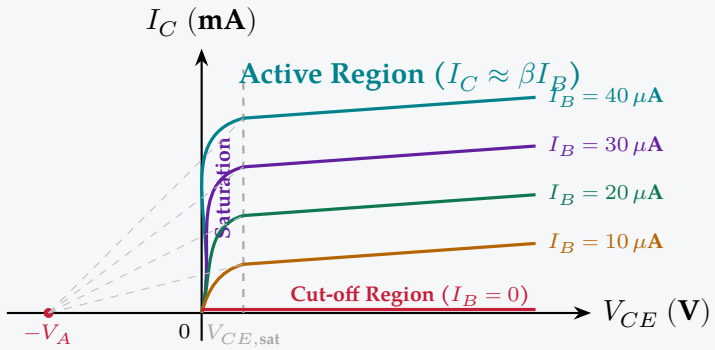
1. **Input Characteristics (I_B vs. V_{BE} for constant V_{CE}):**
   - Resembles the exponential forward I-V curve of a p-n junction diode with knee / cut-in voltage $V_\gamma \approx 0.6$–0.7 V for Silicon.

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- As V_{CE} increases, base width modulation (Early effect) narrows the effective base width W_{eff}, reducing base recombination. Consequently, a higher V_{BE} is required to sustain the same base current I_B, shifting the curve slightly to the right.
- **Dynamic Input Resistance:** $h_{ie} = r_\pi = \left.\frac{\partial V_{BE}}{\partial I_B}\right|_{V_{CE}=\text{const}} \approx 1$ to $5\,\text{k}\Omega$.

2. **Output Characteristics (I_C vs. V_{CE} for constant I_B):** Displays four distinct regions of transistor operation:
   - **Cut-off Region ($I_B \leq 0$):** Both EBJ and CBJ are reverse-biased. The transistor is turned OFF, with only a negligible leakage current flowing: $I_C = I_{CEO} = (\beta + 1)I_{CBO} \approx 0$.
   - **Saturation Region ($0 \leq V_{CE} \leq V_{CE,\text{sat}} \approx 0.2\,\text{V}$):** Both EBJ and CBJ are forward-biased. I_C increases rapidly and linearly with V_{CE}, limited only by external circuit load ($I_{C,\text{sat}} \approx V_{CC}/R_C$), independent of I_B ($I_C < \beta I_B$).
   - **Active Region ($V_{CE} > 0.2\,\text{V}$, $I_B > 0$):** EBJ is forward-biased and CBJ is reverse-biased. The collector current is controlled primarily by the base current ($I_C \approx \beta I_B$). Curves exhibit a slight positive slope due to Early effect, with an output resistance $r_o = \left.\frac{\partial V_{CE}}{\partial I_C}\right|_{I_B} \approx \frac{V_A}{I_C} \approx 40$–$100\,\text{k}\Omega$.
   - **Avalanche Breakdown Region ($V_{CE} > BV_{CEO}$):** The reverse-biased CB junction enters avalanche multiplication, causing I_C to rise steeply and destructively.

Common-Emitter (CE) Output Characteristics (I_C vs. V_{CE})



6. The Ebers-Moll Model

The **Ebers-Moll model** is a generalized, physics-based, large-signal non-linear equivalent circuit model for the BJT valid across **all four operating modes** (Active, Cut-off, Saturation, and Inverted).

6.1 Model Formulation and Equivalent Circuit

The BJT is modeled as two coupled p-n junction diodes connected back-to-back with two current-controlled current sources representing the transistor action:

- An Emitter diode carrying current I_F with a dependent collector current source $\alpha_F I_F$.
- A Collector diode carrying current I_R with a dependent emitter current source $\alpha_R I_R$.

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where:

$$I_F = I_{ES}\left[\exp\left(\frac{qV_{BE}}{kT}\right) - 1\right] \tag{25}$$

$$I_R = I_{CS}\left[\exp\left(\frac{qV_{BC}}{kT}\right) - 1\right] \tag{26}$$

I_{ES} and I_{CS} are the saturation currents of the Emitter-Base and Collector-Base junctions, respectively, and α_F (α_{forward}) and α_R (α_{reverse}) are the forward and reverse common-base current gains.

Ebers-Moll Equations

The terminal currents for an n-p-n transistor are:

$$I_E = I_F - \alpha_R I_R = I_{ES}\left(e^{V_{BE}/V_T} - 1\right) - \alpha_R I_{CS}\left(e^{V_{BC}/V_T} - 1\right) \tag{27}$$

$$I_C = \alpha_F I_F - I_R = \alpha_F I_{ES}\left(e^{V_{BE}/V_T} - 1\right) - I_{CS}\left(e^{V_{BC}/V_T} - 1\right) \tag{28}$$

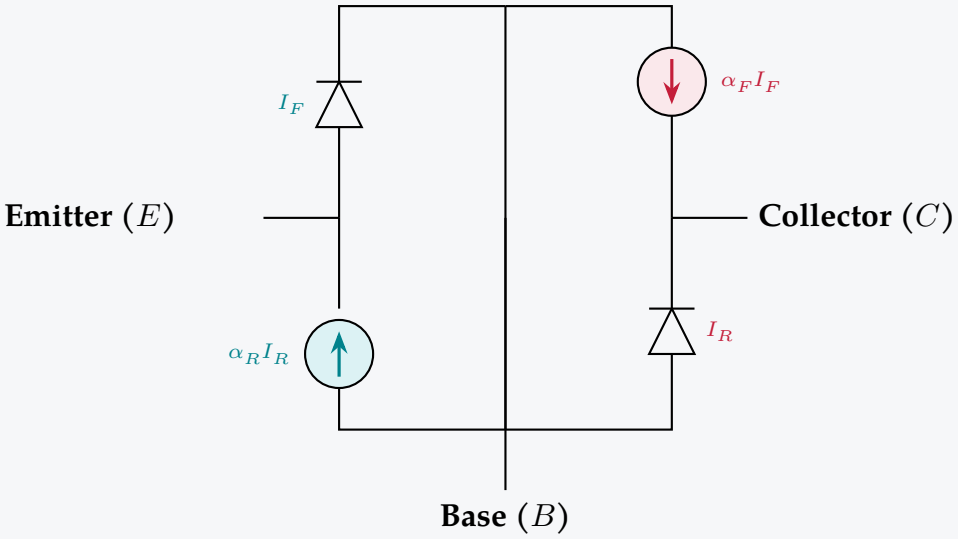
$$I_B = I_E - I_C = (1-\alpha_F)I_F + (1-\alpha_R)I_R \tag{29}$$

The Reciprocity Condition: By thermodynamic reciprocity of linear carrier diffusion across the base:

$$\alpha_F I_{ES} = \alpha_R I_{CS} = I_S \tag{30}$$

where I_S is the fundamental BJT saturation transfer current.

Ebers-Moll Large-Signal Equivalent Circuit (N-P-N Transistor)



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Quick Revision — Module 2

P-N Junction & Bipolar Junction Transistor | EC301

Key Formulas and Metrics at a Glance

Built-in Potential (V_{bi})	$V_{bi} = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$
Depletion Width (W)	$W = \sqrt{\frac{2\epsilon_s}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)(V_{bi}-V_a)}$
Maximum Electric Field	$\mathcal{E}_{\max} = -\frac{2(V_{bi}-V_a)}{W} = -\frac{qN_D x_n}{\epsilon_s}$
Shockley Diode Equation	$I = I_0\left[\exp\left(\frac{V_a}{\eta V_T}\right)-1\right]$
Dynamic AC Resistance (r_d)	$r_d = \frac{\eta V_T}{I_D} \approx \frac{26\,\text{mV}}{I_D}$
Junction Capacitance (C_j)	$C_j = \frac{\epsilon_s A}{W} \propto (V_{bi}+V_R)^{-1/2}$
Diffusion Capacitance (C_d)	$C_d = \frac{\tau_T I_D}{\eta V_T}$
Common-Base Gain (α)	$\alpha = \frac{I_C}{I_E} = \frac{\beta}{\beta+1}$
Common-Emitter Gain (β)	$\beta = \frac{I_C}{I_B} = \frac{\alpha}{1-\alpha}$
Early Effect Output Resistance	$r_o \approx \frac{V_A}{I_C}$
Ebers-Moll Reciprocity	$\alpha_F I_{ES} = \alpha_R I_{CS} = I_S$

One-Line Definitions

- **Depletion Region:** A localized space charge region at the p-n metallurgical interface depleted of mobile carriers.
- **Built-in Potential** (V_{bi})**:** The equilibrium potential difference across the junction maintaining balance between drift and diffusion.
- **Zener Breakdown:** Field-induced quantum tunneling of valence electrons across a very thin depletion barrier under reverse bias.
- **Avalanche Breakdown:** Voltage breakdown caused by cumulative impact ionization multiplying carriers in a wider depletion region.
- **Schottky Diode:** A metal-semiconductor rectifying diode with majority carrier transport and zero reverse recovery storage time.
- **Reverse Recovery Time** (t_{rr})**:** Total time ($t_s + t_t$) required to switch a diode from forward conduction to reverse blocking.
- **Early Effect:** Base-width narrowing under increasing reverse collector-base bias, raising β and output conductance.
- **Ebers-Moll Model:** Large-signal equivalent model coupling forward and reverse diodes to model BJT operation in all four modes.

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Frequently Asked / Viva Questions

Q1. Why does the depletion region penetrate deeper into the lightly doped side of an asymmetric p-n junction?

By the law of space charge neutrality ($qN_Ax_p = qN_Dx_n$), the total uncovered charge per unit area on both sides must be equal. Hence, the depletion width ratio is inversely proportional to doping: $x_n/x_p = N_A/N_D$.

Q2. Why is diffusion capacitance C_d negligible under reverse bias while junction capacitance C_j is negligible under forward bias?

Under reverse bias, forward current is zero, so no minority carriers are injected into the neutral regions ($C_d \propto I_D \approx 0$). Under forward bias, the massive exponential increase in injected minority carriers makes C_d orders of magnitude larger than C_j.

Q3. Compare Zener and Avalanche breakdown on the basis of temperature coefficient.

Zener breakdown has a **negative temperature coefficient** because rising temperature narrows E_g, enhancing tunneling. Avalanche breakdown has a **positive temperature coefficient** because rising temperature increases lattice vibrations, reducing carrier mean free path and requiring higher voltage for collision ionization.

Q4. What are the major advantages of a Schottky diode over a standard p-n junction diode?

- Virtually zero minority carrier storage time ($t_{rr} \approx 0$), enabling ultra-high speed switching (GHz range).
- Lower forward cut-in voltage (0.2–0.3 V vs 0.7 V), resulting in lower conduction power loss.

Q5. Why must the base of a BJT be made extremely thin and lightly doped?

It must be extremely thin ($W_b \ll L_n$) to minimize electron-hole recombination so that almost all injected carriers reach the collector ($I_C \approx I_E$). It must be lightly doped to ensure high emitter injection efficiency ($\gamma \to 1$).

Q6. Explain the physical phenomenon of Early Effect in a BJT.

Increasing V_{CB} widens the collector-base depletion layer, which shrinks the effective neutral base width W_{eff}. This increases the base minority concentration gradient, reduces base recombination, and causes collector current I_C to rise with V_{CE}.

Q7. What is the reciprocity relation in the Ebers-Moll model?

$\alpha_F I_{ES} = \alpha_R I_{CS} = I_S$. It states that the forward and reverse transfer saturation currents across the base are identical, stemming from the reciprocity of minority carrier diffusion across the base.

Q8. What are the conditions on junction biases for a BJT to operate as a closed digital switch (saturation)?

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Frequently Asked / Viva Questions (contd.)

Both the Emitter-Base Junction (EBJ) and the Collector-Base Junction (CBJ) must be forward biased ($V_{BE} \approx 0.7$–$0.8\,\text{V}, V_{BC} \approx 0.5$–$0.6\,\text{V}$), yielding $V_{CE,\text{sat}} \approx 0.2\,\text{V}$.

Q9. How does the reverse saturation current I_0 of a Silicon diode vary with temperature?

I_0 approximately doubles for every 10°C rise in temperature, governed by the exponential temperature dependence of $n_i^2 \propto T^3 \exp(-E_g/kT)$.

Q10. What is the ideality factor η in the diode equation?

η accounts for the dominant recombination/generation mechanism. $\eta \approx 1$ when diffusion current dominates (high forward bias or Germanium), and $\eta \approx 2$ when carrier recombination in the depletion region dominates (low forward bias in Silicon).

Comparison of BJT Configurations (CB, CE, CC)

Parameter	Common Base (CB)	Common Emitter (CE)	Common Collector (CC)
Input / Output Terminals	In: Emitter, Out: Collector	In: Base, Out: Collector	In: Base, Out: Emitter
Current Gain	$\alpha < 1$ (≈ 0.98)	$\beta = \alpha/(1-\alpha)$ (50–300)	$\gamma = \beta + 1$ (High)
Voltage Gain	High ($A_v \gg 1$)	High ($A_v \gg 1$)	Unity ($A_v \approx 1$, buffer)
Input Impedance (Z_{in})	Very low (10–50 Ω)	Moderate (1–5 kΩ)	Very high (100–500 kΩ)
Output Impedance (Z_{out})	Very high (1–2 MΩ)	Moderate (40–100 kΩ)	Very low (20–100 Ω)
Phase Shift (In to Out)	0° (In-phase)	180° (Inverted)	0° (In-phase)
Primary Applications	High-frequency RF, current buffer	General voltage & current amplification	Impedance matching, emitter follower

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